

COURSE SYLLABUS

BACHELOR OF SCIENCE IN INFORMATION SYSTEMS (BSIS) HUMAN COMPUTER INTERACTION

Year & Section	4th Year – Section E
Course	HUMAN COMPUTER INTERACTION
Course Code	HCI143
Program	Bachelor of Science in Information Systems (BSIS)
Year & Section	4th Year – Section E
Course Type	Major / Professional Course
Suggested Term	As prescribed in the approved BSIS curriculum
Credit Units	3 Units
Prerequisite	As prescribed in the approved BSIS curriculum

I. Course Description

This course focuses on the principles and practices of Human Computer Interaction (HCI) and the design of interactive systems that are effective, usable, accessible, and responsive to user needs. Students examine user-centered design, usability, user research, interaction design, interface principles, prototyping, evaluation, accessibility, and basic information architecture. The course applies HCI concepts to the analysis and improvement of information systems and digital interfaces.

II. Course Learning Outcomes

- ILO 1 Explain fundamental concepts, principles, processes, and methods used in Human Computer Interaction.
- ILO 2 Analyze users, tasks, contexts, and requirements to identify appropriate interaction and interface needs.
- ILO 3 Apply user-centered design principles in creating effective, usable, and accessible interfaces.
- ILO 4 Design wireframes, prototypes, navigation structures, and interaction flows for information systems.
- ILO 5 Conduct basic usability evaluation and interpret user feedback to identify interface problems.
- ILO 6 Apply accessibility, ethical, privacy, and inclusive-design considerations in interactive system development.
- ILO 7 Present and document an HCI-based interface solution and justify design decisions using appropriate HCI principles.

III. Weekly Topics and Syllabus Mapping

Week	Topic / Lesson	Learning Activity	Assessment
1	Introduction to Human Computer Interaction	Discuss HCI concepts, history, goals, and relevance to Information Systems	Diagnostic activity / participation
2	Human Factors and User Characteristics	Examine human perception, cognition, memory, attention, and motor abilities	Human factors activity
3	User-Centered Design	Identify users, stakeholders, goals, tasks, and contexts of use	User analysis output
4	User Research and Requirements	Conduct basic interviews, observation, personas, and task analysis	User research / requirements output
5	Interaction Design Principles	Apply consistency, visibility, feedback, affordance, mapping, and error prevention	Design principles activity
6	Information Architecture and Navigation	Create content structures, menus, navigation paths, and task flows	Navigation / task-flow design
7	Interface Design	Develop screen layouts and apply visual hierarchy, typography, and interface conventions	UI design output
8	Midterm HCI Design Evaluation	Present current interface/prototype and review alignment with user requirements	Midterm prototype evaluation
9	Wireframing and Prototyping	Create low- and high-fidelity wireframes or interactive prototypes	Prototype output
10	Usability Testing	Plan and conduct basic usability tests and identify usability problems	Usability test results
11	Accessibility and Inclusive Design	Apply accessibility and inclusive-design considerations to interfaces	Accessibility checklist
12	Mobile and Responsive Interaction	Examine mobile interaction patterns, responsive layouts, and device considerations	Responsive interface activity
13	HCI Evaluation Methods	Apply heuristic evaluation, walkthroughs, surveys, and user feedback techniques	Evaluation report
14	Ethical, Privacy, and Responsible HCI	Consider privacy, ethics, inclusivity, safety, and responsible interface design	Ethics / privacy checklist
15	Interface Refinement and Documentation	Improve the prototype based on evaluation findings and prepare documentation	Revised prototype / documentation
16	Final HCI Presentation and Demonstration	Present, demonstrate, and explain the completed HCI-based interface solution	Final presentation / demonstration

III. Weekly Topics and Syllabus Mapping (continued)

Week	Topic / Lesson	Learning Activity	Assessment
16	Final HCI Presentation and Demonstration	Present, demonstrate, and explain the completed HCI-based interface solution	Final presentation / demonstration

IV. Learning Resources

- Course modules and instructor-provided materials
- HCI and user-centered design books, articles, and reference materials
- UI/UX design and prototyping tools
- Usability testing and evaluation templates
- Accessibility and inclusive-design guidelines
- Institutional and course-specific project guidelines

V. Assessment Methods

Assessment Component	Weight
HCI activities and design exercises	25%
Interface / prototype project development	30%
Usability testing and evaluation	15%
HCI documentation	10%
Accessibility / ethics / design activities	10%
Final presentation and demonstration	10%
TOTAL	100%

VI. General Requirements

- Students are expected to participate actively in lectures, laboratory/design activities, consultations, and collaborative tasks.
- Students must complete assigned HCI outputs and maintain appropriate project files and documentation.
- Students are expected to conduct required usability evaluation, refinement, and presentation activities.
- Students must observe academic integrity and properly acknowledge third-party designs, libraries, templates, references, and other resources used.
- Use of AI-assisted design or development tools must follow instructor and institutional guidelines, and students must be able to explain and demonstrate submitted work.
- Final interface/prototype outputs and supporting documents must be submitted according to the instructor's approved schedule.

VII. References

- Course modules and instructor-provided materials.
- Human Computer Interaction and user-centered design references identified by the instructor.
- Usability, interaction design, information architecture, and interface design references.
- Selected academic articles and reputable materials related to HCI, accessibility, and user experience.
- Institutional guidelines and other resources identified by the instructor.

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BSIS 4th Year – Section E
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